

come with a screen size less than 13.5cm (5.3 inches) and have made PDA devices obsolete. They come with applications and features to manage personal and professional work. The latest hardware capabilities of smartphones also incorporate high computing processors, storage and memory units.

- **Tablets:** These touchscreen devices have screen sizes that vary from 17.8cm to 25.4cm (7 inches to 10 inches). They are mini computers with capabilities like an Internet connection, applications, data storage, video/audio players, etc.
- **Phablets:** These devices generally have screen sizes ranging from 13.5cm to 17.5cm (5.3 inches to 6.9 inches). The word 'phablet' has been coined for devices that are a combination of smartphones and tablets.

Device strategy

It is essential to have fine-grained and well-defined strategies for how devices are used in enterprises. Emerging hardware and software capabilities of Android mobile devices have led to diverse device distribution gateways. Enterprises can select and plan according to their needs and security level enforcement strategies. Each device distribution approach requires a technical support system, application/device management plan, admin console for administration activities, and policies that need to be enforced. Popular enterprise device strategies are BYOD, CYOD and COPE, which are described below.

BYOD: Bring Your Own Device (BYOD) is a popular device strategy with enterprises, wherein users are allowed to access their own personal devices for enterprise applications. As enterprise data and applications are accessed via users' personal devices, this strategy requires well planned and enforced security to protect the data.

CYOD: Choose Your Own Device (CYOD) is a limited version of BYOD. In this device strategy, enterprises can support selected Android devices for application access, and users are requested to choose a device from a proposed device list. This strategy also demands security mechanisms and policies to protect enterprise data on Android devices.

COPE: The Corporate Owned Personally Enabled (COPE) device distribution policy is the way to ensure high levels of security for enterprise data. This is also one of the recommended ways for enterprises, if data is very sensitive, and its loss leads to high risks. Enterprises can use customised COPE Android devices that come with high security mechanisms and limited access for personal usage.

Mobile platform customisation

Android mobile platform customisation is used for fine grained or tailored control of mobile devices. Corporate owned devices can be enterprise-ready, customised Android mobile platform devices. These devices can be managed by an admin console to enforce enterprise policies on devices. Enterprise-ready

customised platforms come embedded with controls for the devices, applications and role based access, which enforce high levels of security management. Android mobile platform customisation can leverage the following aspects:

- Custom branding and UI/UX
- Add or remove device features from the platform
- Integration with third party hardware devices
- Install, remove or upgrade applications on the platform
- Platform porting and upgrades
- Integrate legacy code
- Firmware-over-the-air (FOTA) installation or upgrades

Such mobile platform customisation catalysts can also be leveraged by OEMs (original equipment manufacturers), either for the enterprise or the consumer market.

Mobile applications

Android mobile applications can be broadly divided into two types — system apps and third party apps. System apps are those that come inbuilt with the Android mobile operating system or platforms on the devices, while third party apps are those that are implemented and installed by the developer community.

System apps: Android applications that are implemented with the NDK (native development kit) and can be integrated with mobile platforms or OSs are system apps. These are efficient enough to program applications at hardware level interactions and optimisations like memory, process, execution, etc. Applications provided, by default, from the Android platform on devices are system apps, which are either implemented by the platform community or partner vendor. For example, Android NDK supports systems app implementation via C/C++ programming language, while Android SDK (software development kit) supports native application implementation via the Java programming language.

Third party apps: There are several development approaches for implementing third party Android mobile applications. These can be classified into three types:

- Native apps
- Web apps
- Hybrid apps

Native apps: These are also known as 'thick client' applications, which are implemented via Android mobile device platform native technologies. Here are the imperative characteristics of Android mobile native apps:

- An executable file installs and resides at the mobile device
- Executed directly by the mobile operating system
- Able to use mobile platform or operating system APIs
- Distributed via a platform-specific app store or via an enterprise distribution mechanism

Web apps: These are also known as 'thin client' applications, and are implemented with Web technologies (HTML, CSS and Java Script). Some imperative characteristics for mobile Web apps are:

- Apps are executed by the Android device browser
- Apps can leverage only limited device features for